LAB 3

Component Modelling & Architectural Pattern Selection

Restaurant Table Booking & Pre-Ordering App

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GitHub Repository	https://github.com/SakshamBajpayi/SE-lab-PES1UG24CS404
Based on	Lab 1 requirements and Lab 2 Agile backlog

1. Design Basis

Problem statement. The system allows diners to select an available table using an interactive 2D floor plan, pre-order menu items, calculate kitchen preparation timing, and receive a reservation confirmation. Restaurant managers can update table availability and monitor the floor plan.

- Lab 1 identified five functional requirements: booking, menu & order, preparation timeline, notification, and floor-plan update.
- The key quality constraints are quick table-status synchronization and a responsive mobile floor-plan experience.
- Lab 2 decomposed the same system into eight user stories across four epics, covering booking, menu/pre-ordering, kitchen scheduling, and confirmation/floor management.

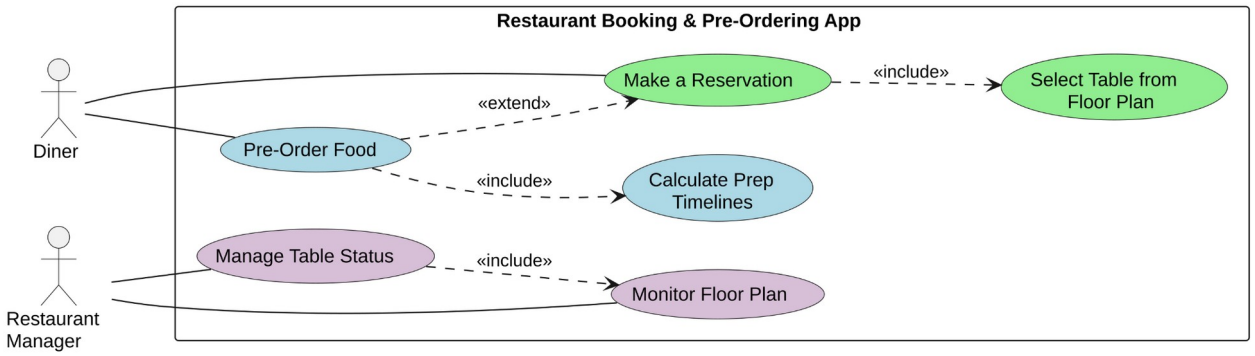


Figure 1. Use-case model carried forward from Lab 1.

2. Component Diagram

The component model decomposes the application into replaceable modules with clear responsibilities and interfaces. The presentation layer contains the two user-facing clients, the application layer holds booking and business logic, and the data/external layer provides persistent storage and notification/status infrastructure.

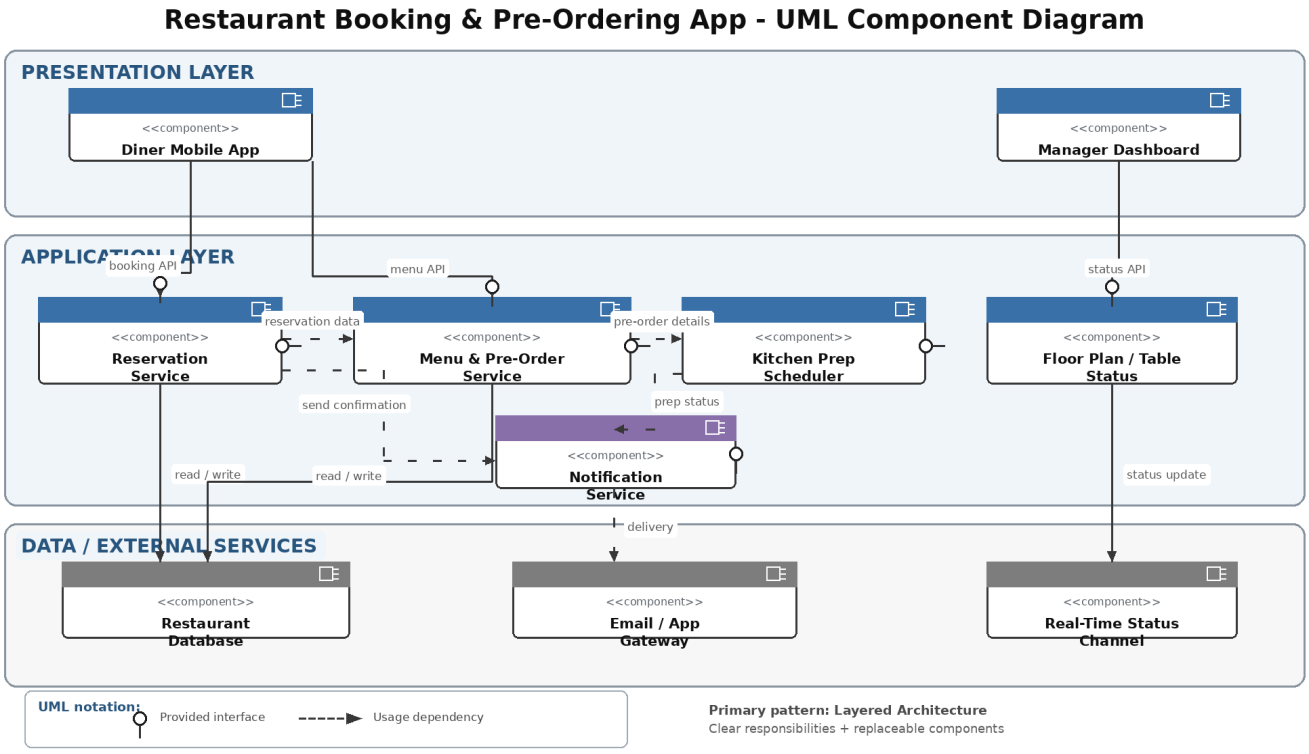


Figure 2. Component diagram for the Restaurant Booking & Pre-Ordering App.

Design note: The diagram uses UML <<component>> notation, provided-interface lollipops and dashed usage dependencies. The interfaces represent the contracts between modules rather than their internal implementation.

3. Component Responsibilities

Component	Main responsibility
Diner Mobile App	Mobile UI for browsing the floor plan, selecting tables, viewing the menu, customizing quantities/notes, and checking booking confirmation.
Manager Dashboard	UI for monitoring the floor plan and changing table availability/status.
Reservation Service	Validates reservation requests, prevents overlapping bookings, stores reservation details, and links a selected table with the booking.
Menu & Pre-Order Service	Manages menu display, quantities and dietary/custom notes; associates pre-orders with reservations.
Kitchen Prep Scheduler	Uses pre-ordered items and reservation time to calculate preparation timing and schedule the kitchen ticket.
Floor Plan / Table Status	Maintains table availability/status and provides the current floor-plan state to diner and manager clients.
Notification Service	Builds reservation confirmations containing table details, pre-order summary and QR/check-in information.
Restaurant Database	Persists tables, reservations, menu selections, preparation schedules and status information.
Email / App Gateway	External delivery point used by the notification component to send the confirmation.
Real-Time Status Channel	Propagates table-status changes so manager views remain synchronized.

4. Key Interfaces & Dependencies

Interface	Provided by	Used by	Purpose
IReservation	**Reservation Service**	Diner Mobile App	Create/check reservation and selected-table details
IOrder	**Menu & Pre-Order Service**	Diner Mobile App	Read menu and submit customized pre-order
IKitchen	**Kitchen Prep Scheduler**	Menu & Pre-Order / Reservation	Request prep timeline and ticket schedule
ITableStatus	**Floor Plan / Table Status**	Manager Dashboard / Diner App	Read and update current table availability
INotification	**Notification Service**	Reservation Service	Send confirmation after successful booking

5. Architectural Pattern Selection

Selected pattern: Layered Architecture

The system is organized into hierarchical layers, with each layer having a specific role and interacting through defined boundaries. This matches the architecture material used in the course, which describes layered systems as a way to separate responsibilities and make individual layers easier to swap or upgrade.

Layer	Components	Role
Presentation	Diner Mobile App; Manager Dashboard	Captures user input and renders the current floor-plan, menu and reservation state.
Application / Business	Reservation; Menu & Pre-Order; Kitchen Scheduler; Floor Plan; Notification	Contains the system rules and use-case logic.
Data / External	Restaurant Database; Email/App Gateway; Real-Time Status Channel	Provides persistence and communication with external delivery/status infrastructure.

6. Architectural Justification

Separation of concerns: Booking, ordering, kitchen scheduling, notifications and floor-plan management are distinct responsibilities, so they can be developed and tested independently.

Clear interfaces: The component boundaries provide explicit contracts such as IReservation, IOrder and ITableStatus. This reduces direct coupling between UI and internal logic.

Performance & synchronization: The floor-plan/status logic is centralized in the application and data layers, supporting the Lab 1 requirement that table-status changes remain synchronized across manager terminals.

Mobile usability: The UI is isolated from business logic, so the mobile floor-plan can be optimized without changing the reservation or database logic.

Maintainability: A change to the notification mechanism, database implementation, or presentation layer can be made with limited impact on the other layers.

7. Why Not the Other Common Patterns?

Pattern	Fit to this system	Reason for selection decision
Client-Server	Applicable because clients request services from a central server.	Useful as a deployment view, but it does not describe the internal separation of booking, ordering, scheduling and data responsibilities as clearly.
Microservices	Can separate business domains into independently deployed services.	More operational structure than needed for this student-scale application; the requirements do not demand independent deployment of each module.
Pipe & Filter	Best suited to a sequence of processing stages.	The app is interaction-heavy and stateful rather than mainly a unidirectional processing pipeline.

8. Requirements-to-Component Traceability

Requirement	Mapped components	Design link
FR-001 Booking	Reservation Service; Floor Plan / Table Status; Restaurant Database	Table selection and reservation are handled through the booking flow.
FR-002 Menu & Order	Menu & Pre-Order Service; Restaurant Database	Supports real-time menu display, quantities and customization notes.
FR-003 Prep Timeline	Kitchen Prep Scheduler; Reservation Service; Menu & Pre-Order Service	Combines reservation time and pre-ordered items to schedule kitchen preparation.
FR-004 Notification	Notification Service; Reservation Service; Email / App Gateway	Creates and delivers confirmation containing table, order summary and QR/check-in details.
FR-005 Floor Plan Update	Floor Plan / Table Status; Manager Dashboard; Restaurant Database; Real-Time Status Channel	Lets the manager block or update tables and keeps status visible to the relevant clients.
NFR-001 Performance & Security	Application / Business + Data layers	Provides a central point for status validation, persistence and synchronization.
NFR-002 Usability	Diner Mobile App; Manager Dashboard	Keeps the interactive floor plan in the presentation layer for responsive mobile/terminal UI.

9. Connection to Lab 2

- SCRUM-5 and SCRUM-6 map mainly to the Floor Plan / Table Status and Reservation Service components.
- SCRUM-7 and SCRUM-8 map to Menu & Pre-Order Service.
- SCRUM-9 and SCRUM-10 map to Kitchen Prep Scheduler and its reservation/order inputs.
- SCRUM-11 maps to Notification Service, while SCRUM-12 maps to Floor Plan / Table Status and the Manager Dashboard.

10. Conclusion

The component model translates the Lab 1 use cases and Lab 2 backlog into a practical software structure. The selected Layered Architecture keeps presentation, business logic and data/external services separated, while the component interfaces make the connections between these modules explicit. This structure directly covers the booking, pre-ordering, kitchen scheduling, notification and floor-management requirements of the restaurant app.

11. References

- Lab 1 - Requirements Analysis & Use-Case Model, Restaurant Table Booking & Pre-Ordering App, Saksham Bajpayi, PES1UG24CS404.
- Lab 2 - Agile Backlog Creation & Sprint Simulation in Jira, Restaurant Table Booking & Pre-Ordering App, Saksham Bajpayi, PES1UG24CS404.
- Software Engineering Unit 2 - Lecture 5 & 6, component diagrams, interfaces, dependencies and architectural patterns.